Shaz (Ethan Shahrooz) Moghaddam

Data Scientist · Python Developer

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Right to work: British citizen · DBS check available on request

PROFILE

Data scientist and Python developer with a Data Science Bootcamp qualification from Imperial College London and HyperionDev (96% average). I take data from raw files to tested, deployed products: cleaning and analysing it, modelling it with scikit-learn and NLP, and shipping it with FastAPI, SQL and Docker. Recent work includes PerilMap, a live climate risk app built on 5 m flood grids, and Bezel Charts, an open-source chart library I publish on npm and PyPI. I'm looking for a data science or ML engineering role where I can own work from the first question to production.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Data and ML: pandas, NumPy, SciPy, scikit-learn, TensorFlow (LSTM forecasting), spaCy, NetworkX, Pydantic

Methods: linear and logistic regression, decision trees, random forests, K-means clustering, anomaly detection (Isolation Forest), causal discovery (PC algorithm), NLP, time-series analysis, evaluation on time-based hold-outs

Geospatial: rasterio, Shapely, pyogrio, raster grids from vector data, MapLibre, Environment Agency and GLA open data

LLMs and AI tools: Claude API for structured extraction and plain-English questions about data; Claude, ChatGPT and Gemini used daily in development

Data and APIs: PostgreSQL, SQLite, SQLAlchemy 2.0, Alembic, FastAPI, Flask, REST APIs, ONS API, OAuth 2.0 (Auth0)

Visualisation: Matplotlib, Seaborn, Plotly Dash, Streamlit, Tableau, React, Next.js, Bezel Charts (my own library)

Engineering: Docker, Git, GitHub Actions (CI and scheduled workflows), pytest, Playwright, Jupyter, Google Cloud Run, Vercel, Render, publishing to npm and PyPI with trusted publishing

SELECTED PROJECTS

PerilMap, Climate Risk for Homes in London

Python · FastAPI · rasterio · Shapely · NumPy · Next.js · TypeScript · MapLibre · Docker · Google Cloud Run · Vercel

- Live web app showing how exposed the area around a London home is to flooding, subsidence and heat: a risk band for each peril with plain-English reasons, the distance to the nearest flood zone, and the zones drawn on a map.
- Turned the Environment Agency's flood geodatabases into 5 m grids, with coverage checks against London's boundary so a gap in the data shows as “No data” and can never look like low risk. Regions are plug-ins, so a new one adds data, not code changes.
- Private by design (searches never stored, locations kept out of URLs and logs, rate limited), WCAG 2.2 AA, and checked by 178 Python and 139 website tests plus Playwright end-to-end tests. Released 1.0.0 in November 2026.

perilmap.vercel.app (live) · github.com/Moghaddam-Labs/perilmap

Bezel Charts, Open-Source Chart Library for JavaScript and Python

JavaScript · Python · Web Components · SVG · npm · PyPI · pytest · Node test runner · Playwright · MIT licence

- Sixteen zero-dependency chart types, from sparklines to waterfalls, box plots and dual-axis combos, as web components on npm and as a Python package on PyPI that draws them from pandas DataFrames for notebooks, reports and READMEs.
- The Python renderers are a line-by-line port, tested byte for byte against the JavaScript across 257 chart cases and 689 number-formatting cases in CI. Ten releases from 1.0 to 1.10, published from Git tags, with keyboard and screen reader support, click events for React and Vue, and per-chart size budgets (8 to 13 kB gzipped).

github.com/ShazMoghaddam/bezel-charts · npmjs.com/package/bezel-charts · pypi.org/project/bezel-charts · Live demo

VoltEdge, Energy Intelligence Platform

FastAPI · Plotly Dash · scikit-learn · NetworkX · SQLite · Docker · Claude API

- Monitors energy use across four industrial sites, with anomaly detection for each site (Isolation Forest), root-cause analysis by causal discovery (PC algorithm), ESG reporting (Scope 2, GRI 302-1, CSRD) and 911 passing tests.
- Evaluated the detector on a 30-day time-based hold-out with labelled anomalies. Found that outlier cleaning was hiding the largest spikes; fixing it raised recall from 76% to 94%, cut false alarms by 58% and doubled F1.

github.com/Moghaddam-Labs/voltedge · Live demo

Jobs & the Economy, Open UK Labour Market Challenge

Python · pandas · Bezel Charts · ONS API · HMRC PAYE payroll data · GitHub Actions · pytest · MIT licence

- Public data challenge on UK jobs, unemployment and pay. One script downloads 33 ONS labour market series and HMRC payroll data, and a report of eight findings is generated from the data, so every number and sentence stays true as new figures arrive.

- Analysis covers long-term sickness (up 620,000 since Covid to 2.74 million), youth unemployment (15.6%), real pay since 2001, a Phillips curve (r = −0.65, 2001 to 2025) and where survey and payroll data diverge. A weekly GitHub Actions job tests the data and opens a pull request after each ONS release.

github.com/ShazMoghaddam/uk-jobs-economy

LedgerLet, Making Tax Digital Platform for UK Landlords

Python · FastAPI · PostgreSQL · SQLAlchemy · Alembic · React · TypeScript · Auth0 · Docker · GitHub Actions

- Full-stack platform that files landlords' quarterly updates to HMRC through its Making Tax Digital APIs, end to end on HMRC's sandbox, over OAuth with encrypted tokens, fraud prevention headers and an append-only audit trail.
- Bank statement CSV import with column mapping and duplicate detection by row fingerprints; money stored as integer pence, with joint-ownership shares applied by one tested rounding rule. Checked by 394 backend tests plus Playwright browser and accessibility tests in CI.

github.com/Moghaddam-Labs/ledgerlet · Live demo

VoltScope, Energy Bill Extraction and Validation

FastAPI · Claude API · Pydantic · SQLite · Docker

- Reads UK commercial energy bills from PDFs and phone photos and extracts them into a fixed Pydantic schema with the Claude API, then runs rule-based checks: MPAN validation, subtotal and VAT reconciliation, estimated readings, duplicate charges and renewal dates. Checked by 50 passing tests based on real anonymised bills.

github.com/Moghaddam-Labs/voltscope

ClinIQ, Clinical Trial Site Performance Dashboard

FastAPI · Plotly Dash · SQLAlchemy · scikit-learn · spaCy · Claude API · Docker

- Compares clinical trial sites on enrolment speed, risk and protocol deviations: groups deviation reports by theme with spaCy, scores site risk with scikit-learn, and answers plain-English questions about trial data with Claude.

github.com/Moghaddam-Labs/cliniq · Live demo

Also: SalaryAxis, salary benchmarking on ONS ASHE data with CPI adjustment and pay gap trends (github.com/Moghaddam-Labs/salaryaxis); and two more open data challenges, on AI adoption in UK business (github.com/ShazMoghaddam/uk-ai-economy) and UK house prices (github.com/ShazMoghaddam/uk-house-prices-economy).

EXPERIENCE

Founder and Data Scientist, Moghaddam Labs 2022 – Present

London (remote) · npm, PyPI and Google Play · Gumroad, Lemon Squeezy and Envato

- Built and shipped the data and AI projects above, and published two open-source libraries: Bezel Charts on npm and PyPI, and Orbs, ten dependency-free loading animations in one 5 kB file (github.com/ShazMoghaddam/orbs).
- Built and published seven Android and Flutter apps on Google Play, owning each one from first idea through to release.
- Sell digital products including finance tools, generative art and screensavers, improving them based on customer feedback.

Career Break Mar 2013 – 2022

Full-time primary carer for my two children

- Project-managed a loft conversion and kitchen extension without a main contractor: council approvals, sourcing and scheduling trades, and the budget. Finished in about 10 months for around £140k, against quotes of £180k to £200k.

Sales Representative (Technology) Mar 2011 – Mar 2013

John Lewis and Currys (via agency) · London

- Advised customers on technology products across several London stores, explaining technical features in plain terms.

EDUCATION AND CERTIFICATIONS

Data Science Bootcamp, Imperial College London and HyperionDev June 2025

96% average. Covered SQL and databases, data cleaning and preprocessing, statistics, visualisation (Tableau and Python), supervised learning, K-means clustering and NLP, finishing with three capstone projects.

The Data Science Course: Complete Data Science Bootcamp (Udemy, 32 hours) Oct 2025

100 Days of Code and The Complete Python Bootcamp (Udemy, 58 hours) Oct 2022 – Jul 2023

ADDITIONAL

Writing: Technical articles on dev.to/shazmoghaddam and seven personal essays on medium.com/@shaz.moghaddam.

Open source: Open-source work selected for the XERJ Corpus Hub (hub.xerj.org/contributors/ShazMoghaddam).

Availability: Full-time, freelance or contract. On-site, hybrid or remote in London, and open to international opportunities.